

AA SL 2023 May P2 TZ0 (no calculator)

1. Consider an arithmetic sequence with $u_1 = 0.6$ and $u_4 = 0.15$.

(a) Find the common difference, d . [2]

The following table shows the probability distribution of a discrete random variable X such that $P(X = n) = \frac{u_n}{k}$, where $n \in \mathbb{Z}^+$, $1 \leq n \leq 4$ and $k \in \mathbb{R}^+$.

n	1	2	3	4
$P(X = n)$	$\frac{0.6}{k}$	$\frac{u_2}{k}$	$\frac{u_3}{k}$	$\frac{0.15}{k}$

(b) Find the value of k . [4]

2. [Maximum mark: 5]

Let a be a constant, where $a > 1$.

(a) Show that $a^2 + \left(\frac{a^2 - 1}{2}\right)^2 = \left(\frac{a^2 + 1}{2}\right)^2$. [3]

Consider a right-angled triangle with sides of length a , $\frac{a^2 - 1}{2}$ and $\frac{a^2 + 1}{2}$.

(b) Find an expression for the area of the triangle in terms of a . [2]

3. [Maximum mark: 7]

The function f is defined by $f(x) = \frac{7x+7}{2x-4}$ for $x \in \mathbb{R}$, $x \neq 2$.

(a) Find the zero of $f(x)$. [2]

(b) For the graph of $y = f(x)$, write down the equation of

(i) the vertical asymptote; [1]

(ii) the horizontal asymptote. [1]

(c) Find $f^{-1}(x)$, the inverse function of $f(x)$. [3]

4. [Maximum mark: 6]

(a) Show that the equation $\cos 2x = \sin x$ can be written in the form
 $2 \sin^2 x + \sin x - 1 = 0$. [1]

(b) Hence, solve $\cos 2x = \sin x$, where $-\pi \leq x \leq \pi$. [5]

5. [Maximum mark: 5]

Find the least positive value of x for which $\cos\left(\frac{x}{2} + \frac{\pi}{3}\right) = \frac{1}{\sqrt{2}}$.

6. [Maximum mark: 6]

Events A and B are such that $P(A) = 0.3$ and $P(B) = 0.8$.

- (a) Determine the value of $P(A \cap B)$ in the case where the events A and B are independent. [1]
- (b) Determine the minimum possible value of $P(A \cap B)$. [3]
- (c) Determine the maximum possible value of $P(A \cap B)$, justifying your answer. [2]

7. [Maximum mark: 4]

Given that $\frac{dy}{dx} = \cos\left(x - \frac{\pi}{4}\right)$ and $y = 2$ when $x = \frac{3\pi}{4}$, find y in terms of x .

8. [Maximum mark: 7]

Consider the functions $f(x) = -(x - h)^2 + 2k$ and $g(x) = e^{x-2} + k$ where $h, k \in \mathbb{R}$.

(a) Find $f'(x)$. [1]

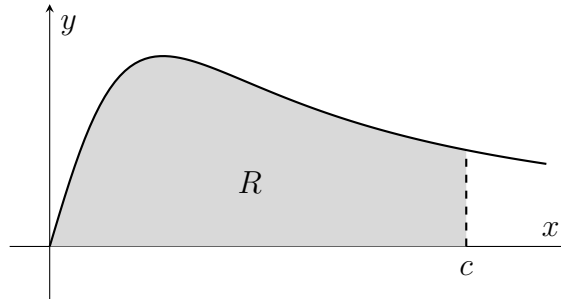
The graphs of f and g have a common tangent at $x = 3$.

(b) Show that $h = \frac{e + 6}{2}$. [3]

(c) Hence, show that $k = e + \frac{e^2}{4}$. [3]

9. [Maximum mark: 6]

The following diagram shows part of the graph of $y = \frac{x}{x^2 + 2}$ for $x \geq 0$.



The shaded region R is bounded by the curve, the x -axis and the line $x = c$.

The area of R is $\ln 3$.

Find the value of c .